

Getting Started with GNU Octave



In the first two parts of this series, we learnt how to use Octave for linear algebraic problems. We also took a quick look at solving simple optimisation problems. Now, we will explore the other features that facilitate numerical and scientific computing in Octave.

Let's delve into Octave's features that relate to statistics, polynomial manipulations and more.

Statistics

Octave has various features for the analysis and interpretation of multi-dimensional numerical data. As is true for most other cases, the data that you want to analyse is usually specified as a matrix, where a row is one observation, and a column indicates the value of the individual components of an observation.

Descriptive statistics

Descriptive statistics give a quantitative measure of the main features of a data set, such as its mean, median, variance, kurtosis and others. Let us now look at the corresponding functions in Octave:

- **mean:** The function to calculate the mean of a data set is `mean(x, dim, opt)`. If `x` is a vector, then it returns the mean of the elements of `x`, i.e.:

$$\frac{1}{N} \sum_{i=1}^N x_i$$

If `x` is a matrix, the default behaviour of Octave is to return a row vector in which each element is the mean of the corresponding column of `x`. For example:

```
octave -3.2.2.93 > x=[1 4 3]
x =
  1  4  3

octave -3.2.2.94 > mean(x)
ans = 2.6667

octave -3.2.2.99 > x=[1 2 3 4 5 6]
x =
  1  2  3
  4  5  6

octave -3.2.2.96 > mean(x)
ans =
 2.5000  3.5000  4.5000
```

The optional argument `opt` may be used to specify the type of mean. The default is *arithmetic mean*, indicated by an 'a'. It is also possible to calculate the *geometric mean*, specified by a 'g' or the *harmonic mean* specified by an 'h'.